SCooP-iT Gym: plan from real data to a validated twin

● validated ◐ implemented, not yet run ○ planned **M** measured from recordings **A** added

1 Real data

two kinds of pass, never mixed

Build passes (site 1)

- ● **Reference map** — SLAM-built, cm-level, checked
- ◐ **Robots, sensors, links** — calibration, paths, RTT/goodput

Held-out passes (another day)

- ○ **Survey + coop passes** — used for nothing but checks
- ○ **Labels on held-out frames** — boxes for scoring step 5

2 Digital twin · Isaac Sim

built from build passes only

- ◐ **Building** — USD from the reference map — **M**
- ◐ **Sensor models** — LiDAR beams, cameras, real mounts — **M**
- ◐ **Paths: recorded** — robots replay where they drove — **M**
- ○ **Paths: new routes** — robot control in Isaac — **A**
- ◐ **Link: logged / fitted** — on recorded paths (scoop_gym) — **M**
- ○ **Link: indexed by place** — survey cells + contention mode — **M**
- ○ **Dropouts, pose error** — fitted from the recordings — **M**
- ◐ **People and objects** — boxes now; assets + semantics next — **A**

Scenario = a setting per part
link: logged | fitted | place · dropouts: fitted · pose: reference | SLAM · people: +N — **× N**

3 Check the twin

against held-out passes; train only on what passes

- ○ **Scans** — ≤ 3 cm from real, same pose
- ○ **Conditions** — fitted dropouts match
- ○ **Link** — in-frame share within 10 pt

4 Train in the twin

- ○ **Perception and comms** — detector, fusion, what to send
- ○ **Motion policies** — e.g. OcclusionHandoff

5 Test on held-out recordings

real data only

- ○ **Perception and comms models** — scored on real frames and the logged link; motion cannot be tested on a recording

6 Deploy on the testbed

- ○ **Motion policies on the real robots** — real outcome vs. twin prediction, across several policies (sim-vs-real correlation)

new recordings refit the twin

held-out passes